Java Programming

CPT111 – Lab Week 10

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CPT111 Java Programming
Lab Week 10 Exercise and Challenge

File, List, Set, Map

Lab Exercise Week #10

- In this week, you will continue coding related to our series on Clock, but uses an ArrayList to store multiple Clock objects
  - A skeleton file and the Clock.java file are given

- In addition, you will solve problems using List and Map

Lab Exercise #10.1 HouseOfClocks 2.0 PrintClocksCollection

- Complete the method printClocksCollection of the class HouseOfClocks
- It prints all the clocks in this HouseOfClock's clock collection
- It will print each clock on a new line with format "Clock " followed by the Clock format HH:MM
- The ArrayList library is imported, there is no need to import other libraries

Lab Exercise #10.2 Same First Letter

- Given a list of non-empty strings, write a method that returns a Map<String, String> with a key for every different first letter seen,
  with the value of all the strings starting with that letter appended with a comma (,) together in the order they appear in the list
- The libraries Arrays, ArrayList, List, Map and HashMap are imported
- Test case 1:

  sameFirstLetter(["alice", "bob", "apple", "banana"]) → {"a": "alice,apple", "b": "bob,banana"}
- Test case 2:

  sameFirstLetter(["after", "all", "this", "time", "always"]) → {"a":"after,all,always", "t": "this,time"}

Lab Challenge Week #10

- You will continue completing HouseOfClocks version with ArrayList

- Additionally, you will complete a problem solving that uses file input/output operations
  - a sample input file Test1File.txt file is given

Lab Challenge #10.1 HouseOfClocks 2.0 numEarlierClocks

- Complete the method numEarlierClocks in the class HouseOfClocks
- It returns the number of clocks in this HouseOfClock's clock collection that is earlier than that Clock

- The ArrayList library is imported, there is no need to import other libraries

Lab Challenge #10.2 Count Vowel File

- Complete the Java method int countVowelFile(String filename) that given a file name of a text file, return the number of vowels in the text file

- Import setting in the system:
  import java.io.*;
  import java.util.*;

- Test case:
  countVowelFile("TestFile1.txt") → 4

Thank you for your hard work!

- If you finish early, feel free to use the remaining time to work on other online exercises or CW3

- You may leave the lab once you have complete the Lab 9 Challenge and it has been ASSESSED by a TA !
  - Please note that the **Lab Challenge #10.1 and #10.2 are graded** and contributes to your CW2 marks